

8/2/2025

- 8a) ~~TAP~~ : a) To construct a binary search tree. b) To traverse the tree using all the methods i.e, in-order, pre-order and post-order c) To display the elements in the tree.

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
struct node {  
    int data;  
    struct node *left, *right;  
};
```

```
struct node *createnode (int value) {  
    struct node *newnode = (struct node *) malloc (Size of (struct node));  
    newnode->data = value;  
    newnode->left = newnode->right = NULL;  
    return newnode;  
}
```

```
struct node *insert (struct node *root, int value) {  
    if (root == NULL) {  
        return (createnode (value));  
    }  
    if (value < root->data) {  
        root->left = insert (root->left, value);  
    }  
    else if (value > root->data) {  
        root->right = insert (root->right, value);  
    }  
}
```


}

return root;

{

void inorder (struct node * root) {

if (root == NULL) {

return;

}

inorder (root → left);

printf ("%d ", root → data);

inorder (root → right);

~~preorder~~

{

void preorder (struct node * root) {

if (root == NULL) {

return;

}

printf ("%d ", root → data);

preorder (root → left);

preorder (root → right);

}

void postorder (struct node * root) {

if (root == NULL) {

return;

}

postorder (root → left);

postorder (root → right);

printf ("%d ", root → data);

{

void display (struct node * root) {

printf ("BST elements (Inorder): ");

inorder (root);

printf ("\n");

}

int main () {

struct node * root = NULL;


```

int choice, value;
while (1) {
    printf("\n --- Binary Search Tree Menu --- \n");
    printf("1. Insert into BST \n");
    printf("2. Inorder Traversal \n");
    printf("3. Preorder Traversal \n");
    printf("4. Postorder Traversal \n");
    printf("5. Display BST \n");
    printf("6. Exit \n");
    printf("Enter choice: ");
    scanf("%d", &choice);
    switch (choice) {
        case 1:
            printf("Enter value to insert: ");
            scanf("%d", &value);
            root = insert(root, value);
            break;
        case 2:
            printf("Inorder Traversal: ");
            inorder(root);
            printf("\n");
            break;
        case 3:
            printf("Preorder Traversal: ");
            preorder(root);
            printf("\n");
            break;
        case 4:
            printf("Postorder Traversal: ");
            postorder(root);
            printf("\n");
            break;
        case 5:

```



```

        display(root);
        break;
    }
    case 6:
        exit(0);
    default:
        printf("Invalid choice! Try again.\n");
}
return 0;
}

```

Output:

--- Binary Search Tree Menu ---
 1. Insert into BST
 2. Inorder Traversal
 3. Preorder Traversal
 4. Postorder Traversal
 5. Display BST
 6. Exit

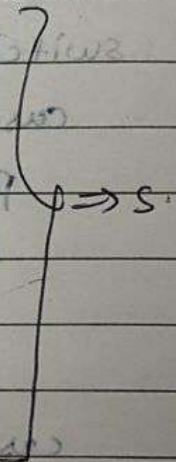
Enter choice: 1
 Enter value to insert: 50
 S

Enter choice: 1
 Enter value to insert: 10
 S

Enter choice: 1
 Enter value to insert: 30
 S

Enter choice: 2
 Inorder Traversal: 10 30 50
 S

Enter choice: 3
 Preorder Traversal: 50 10 30
 S



Enter choice: 4.

Postorder Traversal: 30 20 50

S

Enter choice: 5

~~BST elements (Inorder): 10 30 50~~

~~S~~

~~Enter choice: 6.~~

~~Q~~
8/11/25
Saur